

GREEN INDUSTRIES

Green industries courses offer students opportunities to investigate how to care for and sustainably manage our natural and living resources. Students will explore a variety of areas within green industries, including agriculture, floristry, forestry, horticulture, and/or landscaping.

Students will gain practical skills and knowledge by completing a variety of authentic, industry-relevant activities and projects, such as plant propagation and greenhouse maintenance activities; landscape design and planning exercises; assignments that include farm or forestry management plans; and various construction projects. In addition, students will be introduced to concepts pertaining to biodiversity, environmental sustainability, and natural versus artificial products. Students will also investigate the numerous and varied ways in which green industries and society are interdependent.

Green industries courses will prepare students for working safely in the many fields that are encompassed in this subject area. Students will learn about career opportunities in the green industries and about the Essential Skills and work habits that are important for success in these fields.

The list of approved emphasis areas for green industries can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Green Industries, Grade 12

University/College Preparation

THJ4M

This course focuses on more complex concepts and skills related to the green industries. Students will focus on developing process skills, design and management techniques, and ways of enhancing environmental sustainability. They will also examine social and economic issues related to the green industries, learn about safe and healthy working practices, study industry standards and codes, and explore career opportunities. The knowledge and skills acquired in this course will prepare students for more specialized studies at the college and university level.

Prerequisite: Green Industries, Grade 11, University/College Preparation

A. GREEN INDUSTRY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of species classification and identification and explain relationships between species and geographical regions;
- A2.** analyse the effects of biotic and abiotic factors on growth and post-harvest quality;
- A3.** develop and evaluate designs or processes for a variety of applications in the green industries;
- A4.** use mathematical, documentation, research, and communication skills as they apply to the green industries.

SPECIFIC EXPECTATIONS

A1. Species Classification and Geographical Regions

By the end of this course, students will:

- A1.1** distinguish between different plant and/or animal groups on the basis of key identification characteristics (*e.g., native and non-native species, dairy and beef cattle, deciduous and coniferous shrubs, monocotyledonous and dicotyledonous plants*), and identify species using both common and scientific names (*e.g., white birch [also known as paper birch or canoe birch] [Betula papyrifera] and Mountain paper birch [Betula cordifolia], euonymus [gen. Euonymus] and Emerald Gaiety [Euonymus fortunei 'Emerald Gaiety']*);
- A1.2** identify geographical regions in Canada on the basis of classification criteria relevant to the green industries (*e.g., plant hardiness, growing degree days, elevation, soil type, soil moisture*), and explain how geographical factors determine the distribution of species in these regions;
- A1.3** compare different kinds of ecosystems in terms of their biodiversity (*e.g., a climax forest versus a rejuvenated forest, natural versus managed land, a cultivated field versus a greenhouse*), and explain how biodiversity affects the stability of ecosystems (*e.g., monocultures versus diversified ecosystems*).

A2. Factors Affecting Growth and Product Quality

By the end of this course, students will:

- A2.1** analyse the effects of abiotic factors on growth and post-harvest quality (*e.g., effects of differences in soil composition, climate, water quality and quantity, topography*);
- A2.2** analyse the effects of biotic factors on growth and post-harvest quality (*e.g., physiological effects of pests and diseases, invasive species, genetic variations*);
- A2.3** assess the effects of interactions between abiotic, biotic, and cultural factors on a variety of ecosystems (*e.g., forests in various stages of natural succession, golf courses, fish farms, organic farms, riparian zones*);
- A2.4** compare the effectiveness of different integrated pest management techniques for a variety of applications (*e.g., cultural [tilling and mulching], physical [crop rotation], environmental [introduction of beneficial insects], biological [fungi, nutrients], chemical [pheromones, chemical pesticides]*).

A3. Designs and Processes

By the end of this course, students will:

- A3.1** explain the steps required to create designs or plans for a variety of applications in the green industries (*e.g., timber cruising, surveying, perennial gardens, farms, environmental assessments*);

A3.2 explain advanced systems, processes, and techniques relating to the propagation, maintenance, and care of plants or animals (*e.g., irrigation systems, tree support and protection systems, plantation tending, prescribed burning, regeneration*);

A3.3 evaluate the appropriateness and effectiveness of a management process (*e.g., environmental impact assessment, tree or crop loss assessment, herd health evaluation, growth and yield monitoring*);

A3.4 describe how advanced technologies and scientific knowledge are being applied to processes in the green industries (*e.g., geographical information systems, global positioning devices, three-dimensional CAD applications, aerial and satellite images, computerized feeding systems, computerized inventory systems, genetic modification*), and assess their impacts.

A4. Technological and Mathematical Literacy and Communication Skills

By the end of this course, students will:

A4.1 demonstrate an understanding of terminology used in the green industries and use it correctly in oral and written communication (*e.g., biodiversity, tendril, balance, pergola, tilth*);

A4.2 apply effective documentation practices to a variety of procedures related to green industry operations (*e.g., record data from timber cruises; maintain production and maintenance records; document Hazard Analysis and Critical Control Point [HACCP] activities; prepare inventories, estimates, and invoices*);

A4.3 interpret charts, graphs, aerial photographs, maps, and other visual presentations of information used in the green industries;

A4.4 analyse commercial or technical issues of current significance in the green industries, using information from media and industry sources (*e.g., marketing boards, industry organizations, commodity websites, trade publications, news reports*);

A4.5 accurately perform complex measurements and calculations required for advanced applications in the green industries (*e.g., basal area prism sweeps, pricing, surveys, economic forecasts, grade and slope measurements*).

B. GREEN INDUSTRY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate an understanding of and apply design and production practices that are commonly used in the green industries;
- B2.** develop and apply management strategies for assessing and controlling biotic, abiotic, and cultural factors that affect plant and/or animal quality;
- B3.** demonstrate competence in technical skills relating to specific applications and tasks within the green industries.

SPECIFIC EXPECTATIONS

B1. Design and Production

By the end of this course, students will:

- B1.1** develop a design and/or process that fulfills a specific functional or aesthetic requirement (e.g., a graphic design, a barn design, a specialty garden design, an urban forest regeneration schedule, an advertising brochure; specialty pruning techniques);
- B1.2** design and implement a management plan or site layout for a specific application (e.g., a natural disturbance response and restoration plan, a site survey and construction implementation plan for a landscape design, a growing system for plant production and distribution, plant selection and schedule for crop rotation, animal housing, an urban forest development plan);
- B1.3** demonstrate competence in the use of biological techniques for propagating and maintaining a variety of species (e.g., cone selection, transplanting large trees, reforestation, insect control, hybridization, grafting, artificial insemination);
- B1.4** create plans or designs for green industry projects that enhance biodiversity (e.g., moisture conservation, xeriscaping, integrating diverse native plants, sustainable water gardening).

B2. Plant and Animal Management Strategies

By the end of this course, students will:

- B2.1** analyse biotic conditions affecting the health of plants and/or animals and the quality of products derived from them, using a variety

of diagnostic procedures (e.g., pest counts, pest determination, microscopic investigation, visual inspection, blood testing, cavity assessment);

- B2.2** analyse abiotic conditions affecting the health of plants and/or animals and the quality of products derived from them, using a variety of diagnostic procedures (e.g., nutrient balance testing, soil and water testing, form defect analysis, air quality assessment);
- B2.3** apply a variety of pest and disease control techniques (e.g., integrated pest management, crop rotation, animal inoculation, instituting invasive species controls), and assess their effects on plant and/or animal stock and the environment;
- B2.4** develop and apply best management practices for enhancing environmental sustainability within the green industries (e.g., herd management, native species selection and placement, forest certification, cut selection, local purchasing, composting, integrated pest management, water management, biogas production from wastes).

B3. Technical Skills

By the end of this course, students will:

- B3.1** demonstrate competence in the technical skills required to complete a variety of projects or processes in the green industries (e.g., using or programming a computerized climate control system, measuring trees, grading land, formulating feed);
- B3.2** build a variety of structures that are commonly used in the green industries (e.g., roads, skid trails, pens, pergolas, retaining walls, brick pavements);

B3.3 demonstrate competence in related technical skills required to complete projects within the green industries (*e.g., welding, wiring, making electrical repairs, plumbing, small-engine operation and maintenance, graphic communication, veterinary procedures such as administering medications*);

B3.4 demonstrate competence in specialized technical skills required for unique applications (*e.g., installing low-voltage lighting, working at heights, xeriscaping, topiary shaping, using specialized software applications*).

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** assess options for achieving environmental sustainability in green industry operations;
- C2.** analyse social, economic, and cultural relationships involving the green industries.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** evaluate green industry operations and processes in terms of their impacts on environmental sustainability (*e.g., global floral sourcing and greenhouse gas emissions, by-product management and water quality, monocultures and biodiversity, genetically modified products and effects on pesticide use and biodiversity*);
- C1.2** analyse ways of reducing negative or enhancing positive environmental consequences through the use of particular materials, products, processes, and disposal methods (*e.g., nutrient recycling, spot spraying for insects and fungus, using organic fertilizer, composting, xeriscaping*);
- C1.3** describe methods used in the green industries to balance economic sustainability with environmental responsibilities (*e.g., selective breeding, selective cutting, organic production methods, restricted cattle crossings and buffer zones to prevent water contamination, environmental best management practices*);
- C1.4** describe the benefits of alternative practices that reduce the environmental impact of green industry operations (*e.g., living walls, natural-scaping, xeriscaping, forest certification, tree marking guidelines, fibre crops, armatures and grid work, alternative animal housing systems*);

- C1.5** describe legislation, regulations, standards, and guidelines relating to environmental protection that affect operations in the green industries. (*e.g. Greenbelt Act, Fisheries Act, Crown Forest Sustainability Act, Nutrient Management Act, Forest Fires Prevention Act, pest control regulations*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** analyse ways in which the green industries are affected by the social, economic, and cultural characteristics of the communities in which they operate (*e.g., relations with Aboriginal communities, ethnic preferences and demands for specialized food products, income distribution and demand for landscaping services and horticultural products, age profile of community and requirements for different types of outdoor recreation facilities*);
- C2.2** assess the economic importance of linkages between the green industries and related industries and technologies (*e.g., agriculture: food processing industry, farm implement industry; horticulture: shipping industry, event-related businesses [funeral homes, wedding planners]; landscaping: recreational industries, small-engine industry; forestry: heavy equipment industry, paper-consuming industries such as newspapers*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and apply safe working practices as they relate to the green industries;
- D2.** demonstrate an understanding of the business and regulatory environment of the green industries;
- D3.** identify careers in the green industries that require postsecondary education, and describe the qualifications required for entry into these occupations.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** describe common industry hazards (e.g., *ergonomic hazards, mechanical hazards, chemical hazards*) and sources of information about accident prevention (e.g., *Workplace Hazardous Materials Information System [WHMIS], Passport to Safety*);
- D1.2** identify the personal protective clothing and equipment needed to perform various green industry tasks safely, and use as required to ensure their own and others' safety in the work environment (e.g., *safety glasses, hard hats, safety boots, fall protection gear, equipment guards*);
- D1.3** demonstrate the ability to make appropriate safety decisions for personnel on the basis of environmental and site conditions (e.g., *weather conditions, presence of poisonous plants or dangerous gases, hazardous trees, reliability of communications in remote areas, access to emergency services*) and level of crew training and experience;
- D1.4** demonstrate an understanding of and apply safe procedures for using and maintaining materials, tools, and equipment (e.g., *avoid moving parts and pinch points; perform a circle check of vehicles and equipment; check condition of materials, hydraulic systems, and protective equipment; check oil and fuel levels*);
- D1.5** demonstrate an understanding of and apply safe procedures for handling plants and/or animals (e.g., *dethorning plants, using ergonomic lifting techniques or devices, using hand protection, securing loads correctly for transport, understanding animal perception, avoiding actions that startle animals, using chutes and restraining devices*);

- D1.6** demonstrate an understanding of specific components of the Occupational Health and Safety Act (e.g., *use of hazardous materials in the workplace, duties of employers, rights and responsibilities of workers*).

D2. Business and Regulatory Environment

By the end of this course, students will:

- D2.1** identify industry associations, government departments, and non-governmental organizations that are involved with matters that affect the green industries (e.g., *local growers' associations; provincial and federal agriculture, health, environment, and resource departments; environmental NGOs*);
- D2.2** explain economic principles, concepts, and arrangements that affect operations in the green industries (e.g., *transportation routes, bonuses, quota system, free trade, treaties, supply and demand*);
- D2.3** outline and comply with legislation and guidelines governing the quality and safety of green industry products and services (e.g., *product quality regulations, grading standards, inspection requirements, voluntary guidelines such as the Landscape Ontario guidelines for the landscaping industry*).

D3. Career Opportunities

By the end of this course, students will:

- D3.1** identify careers in the green industries that require postsecondary education, and describe the nature of the work in these occupations;

- D3.2** describe the educational programs, training, and certification needed for entry into a variety of green industry occupations requiring post-secondary education;
- D3.3** investigate areas of specialization within the green industries (*e.g., lighting systems, water features, irrigation systems, GIS analysis, robotics, automation, entomology, pathology, tissue culture, agronomy, marketing, environmental management, farm management*);
- D3.4** identify the scope and strength of the labour force within the green industries regionally, nationally, and internationally;
- D3.5** demonstrate an understanding of and apply the Essential Skills that are important for success in the green industries, as identified in the Ontario Skills Passport (*e.g., reading text, writing, document use, computer use, oral communication, numeracy, thinking skills*);
- D3.6** demonstrate an understanding of and apply the work habits that are important for success in the green industries, as identified in the Ontario Skills Passport (*e.g., initiative, organization, accountability, ethical conduct*);
- D3.7** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in areas related to the green industries (*e.g., Passport to Safety certificate, technical reports, competency checklists, Ontario Skills Passport Work Plan*), and explain why having a current portfolio is important for career development and advancement.